

# An adaptive hybrid model for wildfire front forecasting based on cellular automata, multi-agent UAV observations, and binary data assimilation: A case study of the 2021 Dixie Fire

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**Short title:** Hybrid CA+MAS wildfire front forecasting

## Abstract

This study presents a retrospective case-study evaluation of a hybrid framework for daily wildfire-front forecasting during the 2021 Dixie Fire in California, USA. The framework couples a stochastic cellular automaton (CA) with a multi-agent system (MAS) of simulated UAV observations and a lightweight binary data-assimilation scheme. The model uses topography, vegetation, fuel proxies, and meteorological forcings derived from Copernicus GLO-30/SRTM, ESA WorldCover, MODIS NDVI, MODIS burned-area products, VIIRS active-fire detections, and ERA5-Land data. The CA produces a daily forecast of wildfire evolution, while the MAS directs a configurable set of simulated aerial agents toward spatially distributed high-information regions identified from predicted growth and fire-front structure. Agent observations are assimilated through a binary confirm-deny update that suppresses false ignitions and restores missed burning cells before the next forecast step; the observed footprint can also support local blending of daily weather fields before propagation resumes. In the verified evaluation window from 14 to 18 August 2021, with corrected road and water barrier masks enabled in the CA component, the hybrid CA+MAS framework increased the mean Intersection over Union (IoU) from 0.6664 for the standalone baseline CA to 0.6992, corresponding to an absolute gain of 0.0328, while the mean F1 score increased from 0.7998 to 0.8229. These findings indicate that targeted recurrent aerial observations can improve short-horizon wildfire-front reconstruction while preserving the computational simplicity and interpretability of the underlying cellular automaton. **Keywords:** wildfire front forecasting, wildfire spread, cellular automaton, multi-agent system, unmanned aerial vehicle, data assimilation

# Introduction

Wildfires are increasingly recognized as a systemic environmental and societal risk because they threaten public health, critical infrastructure, ecosystems, and evacuation safety. Recent studies indicate that climate-driven drought, heat extremes, and strong winds are lengthening fire-season duration and expanding the spatial extent of fire-prone conditions, which increases the need for timely and interpretable forecasting tools [1, 2].

These demands are especially acute in the wildland-urban interface (WUI), where rapidly changing wildfire front behavior can affect exposure, suppression priorities, and evacuation planning. However, strong wind shifts, heterogeneous fuel conditions, and local topographic effects are not unique to the WUI; they complicate wildfire spread forecasting more broadly and make static fire-spread estimates quickly outdated.

Existing wildfire modeling approaches address different parts of this problem but rarely combine speed, physical interpretability, and adaptive observation. Fire Area Simulator (FARSITE) and Prometheus, the Canadian Wildland Fire Growth Simulation Model, provide interpretable fire-spread simulations, but they require careful data preparation and can be demanding in operational settings [3, 4]. Remote-sensing and machine-learning approaches can extract useful spatiotemporal patterns from satellite products, but they often depend on large training datasets and may offer limited interpretability in rapidly evolving wildfire environments [5, 6].

Observation-informed wildfire forecasting has also been explored using satellite-derived fire progression products and data-assimilation schemes that update model states with newly available observations [7–9]. These studies support the value of closed-loop forecast updating, but they do not remove the practical need for computationally lightweight and interpretable forecasting frameworks.

This gap motivates a closed-loop forecasting framework in which a physically interpretable spread model is updated with targeted observations. In this study, the forecasting target is daily wildfire front progression. We couple a stochastic cellular automaton (CA) with a multi-agent system (MAS) of simulated unmanned aerial vehicle (UAV) observers that are directed toward informative regions of the predicted fire front. The CA represents the daily evolution of the wildfire front as a function of terrain, vegetation, fuel proxies, and meteorological forcing, while the MAS supplies localized observations that are assimilated before the next forecast step.

The empirical framework combines satellite-derived observational products with reanalysis forcing data at a daily simulation time step. MODIS burned-area and VIIRS active-fire products are used as observational constraints and evaluation references, whereas ERA5-Land is used as meteorological forcing rather than as direct fire observation. Static spatial inputs include topography, land-cover-derived fuel proxies, vegetation condition, and barrier layers.

Rather than claiming a universal operational system, this work presents a retrospective case-study evaluation of an interpretable hybrid CA+MAS framework for the 2021 Dixie Fire in California, USA. The novelty lies in combining an explicit daily wildfire-front CA forecast, informative-region UAV tasking, and a lightweight binary assimilation rule within one reproducible framework. The following sections review related approaches, describe the data and methodology, and evaluate the framework against observed wildfire dynamics.

## Overview of existing approaches and their limitations

### Cellular automata and physics-based wildfire spread models

One of the most common approaches to modeling the spatial dynamics of forest fires is cellular automata (CA), in which the landscape is represented as a grid of cells that transition between discrete states—*burning*, *burned*, and *unburned*—depending on the state of neighboring cells and a defined set of rules. This method allows consideration of topography, fuel combustibility, wind, and other spatial factors, providing a balance between computational simplicity and realism.

An example of a modern solution is HexFire, which uses a hexagonal grid in which all nearest neighbors are equidistant, reducing the anisotropy of fire spread. The model requires only a small parameter set and a few core input maps, such as fuel combustibility, wind speed, and wind direction, while simulating spread

through both direct contact and ember transport [10]. Cellular automata can be deterministic or stochastic, and adaptive cellular-automaton variants have also been used to represent abrupt spread changes under strong wind conditions [11].

Physics-based models, particularly advection–diffusion–reaction (ADR) formulations, provide a more detailed description of combustion processes based on conservation laws for energy and mass. These models account for fuel moisture, radiation, topography, and air movement [12]. In this family of approaches, ignition and spread depend on interacting physical terms, whereas radiative effects are commonly represented through nonlinear temperature-dependent transport terms.

Related coupled-model studies also examine wildfire spread near populated areas, where interactions among spread dynamics, built environments, and suppression constraints become especially important [13].

## Mapping combustible materials using artificial intelligence

Accurate assessment of the types and distribution of combustible materials is a critical component of fire modeling and forecasting. The integration of remote-sensing data with machine-learning algorithms plays a key role in this process.

A modern example is the FuelVision system, which uses multispectral, radar, and topographic predictors to construct fuel maps [2]. In that line of work, ensemble learning and related data-fusion strategies are used to improve mapping robustness when field observations are sparse or class distributions are imbalanced. Related studies also note that synthetic-sample generation and post-processing with vegetation indices can help stabilize fuel-class mapping when field reference data are limited, especially for distinguishing non-burnable classes from combustible vegetation.

More broadly, combining remote-sensing products with modern machine-learning workflows can improve the reliability and scalability of combustible-material mapping, thereby providing a stronger input base for physical, mathematical, and stochastic fire-spread models.

## Deep learning-based fire spread prediction

Deep-learning methods have expanded the range of data-driven spatiotemporal wildfire-prediction approaches.

Convolutional neural networks (CNNs) are used to extract spatial features from satellite data, enabling the detection of active fires and the direction of the fire front. Multi-kernel CNN designs have been used to better account for landscape heterogeneity in next-day spread prediction tasks [14].

Review literature also reports recurrent and attention-based models for capturing temporal fire evolution and longer-range spatial dependencies in satellite-driven wildfire prediction tasks [6]. In particular, ConvLSTM-style architectures and hybrid attention-augmented recurrent models are used to combine spatial fire patterns with short-term temporal evolution, thereby extending purely frame-based prediction toward sequence-aware spread forecasting.

Transformer architectures have also been explored to model longer-range spatiotemporal dependencies in wildfire prediction [6].

## Multi-agent systems and autonomous control

Modern forest fire monitoring and suppression tasks require not only accurate models but also rapid, coordinated action. This is where multi-agent systems (MAS), reinforcement learning, and multi-agent reinforcement learning play an important role.

Recent reviews highlight that autonomous UAV platforms can support fire detection, mapping, navigation, and decision support, while conceptual drone-swarm systems have been proposed for coordinated suppression and front monitoring in rapidly evolving wildfire environments [15, 16].

Related benchmark and simulation studies also explore coordinated multi-agent wildfire response in more explicitly heterogeneous settings, including scenarios with firefighters, aerial agents, and ground equipment trained or evaluated under shared resource constraints [15, 17].

In addition to software simulators, physical platforms based on the Internet of Things and UAVs are being actively developed for early detection and targeted firefighting [17]. These integrated sensing architectures are relevant here because they emphasize rapid observation, communication, and local situational updating rather than stand-alone spread prediction alone.

Other studies discuss swarm-coordination strategies such as belief-space planning for distributed autonomous sensing [16]. Although these methods are not implemented in the present manuscript, they help frame the broader literature context in which coordinated observation and task allocation are treated as coupled problems.

Taken together, the reviewed studies indicate a persistent trade-off: no single family of methods simultaneously provides interpretable fire spread dynamics, efficient computation over large domains, and adaptive state updating under rapidly changing conditions [6, 10–12, 14–17].

Models based on cellular automata (CA) are easy to implement and can reproduce the spatial dynamics of a fire, considering local factors such as topography, fuel, and wind [10, 11], but they are sensitive to parameter settings and lack built-in mechanisms for real-time data updates, limiting their applicability in operational scenarios. Physical-mathematical advection-diffusion-reaction (ADR) models provide a more detailed description of combustion processes, including fuel moisture, radiation, and air movement [12], but their practical use in online forecasting is limited by high computational complexity and reliance on numerous input parameters. Deep learning models have shown strong performance for predicting fire spread based on satellite data, particularly when employing CNNs, convolutional long short-term memory (ConvLSTM) networks, and transformer architectures [6, 14], but these models require substantial amounts of labeled data, are not always physically interpretable, and often fail to account for dynamic real-world environmental factors. Multi-agent systems (MAS) and reinforcement-learning-based solutions show promise for autonomous fire response and suppression tasks [15–17]. Scenarios involving heterogeneous agents such as drones, bulldozers, and firefighters enable the simulation of coordinated collective actions. Nevertheless, the effectiveness of these systems is largely determined by the accuracy of the underlying environmental model and the reliability of input data [17].

This gap motivates the hybrid computational architecture adopted in the present study. The CA component serves as the base spread model, simulating fire propagation while accounting for topography, vegetation, humidity, and meteorological factors, thereby preserving physical interpretability and stability [10]. The MAS component is represented here by a simulated UAV observation layer that supplies targeted observations for retrospective data assimilation rather than by real deployed aircraft. The resulting closed loop of forecast, target selection, observation, and state updating is designed to retain the transparency of CA-based wildfire modeling while adding the adaptive observation logic that is often missing from stand-alone spread models.

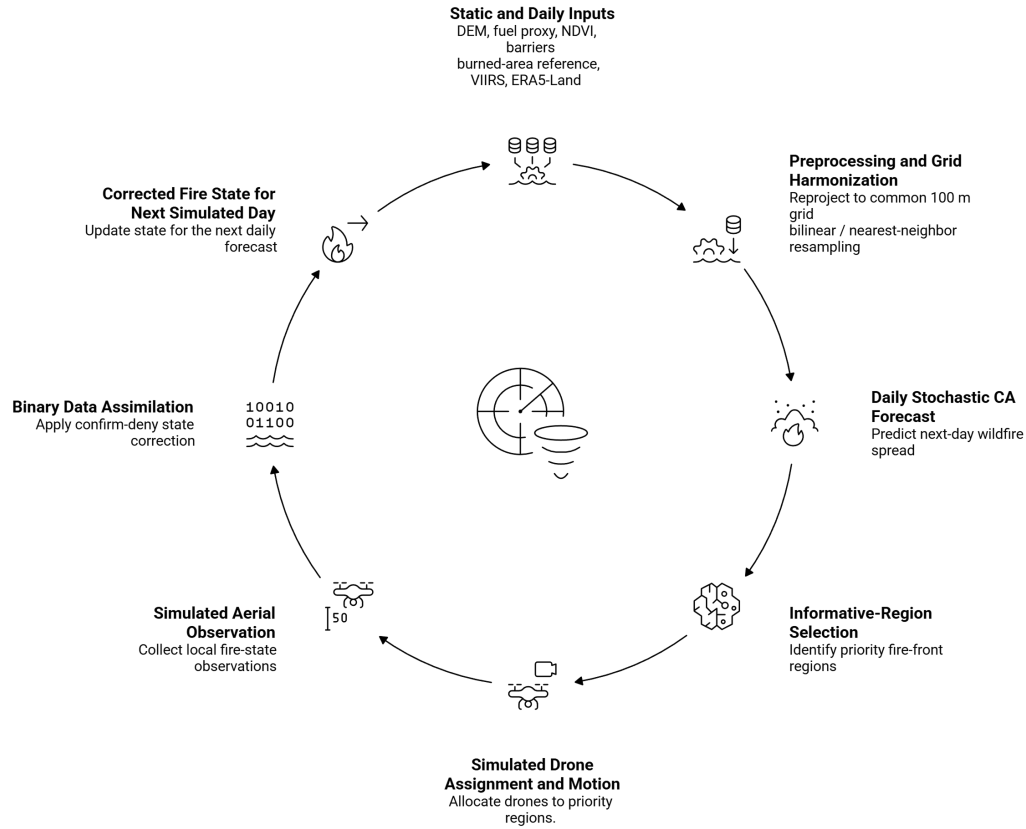
## Materials and methods

The proposed integrated framework combines a cellular automaton for daily fire-front propagation with a simulated drone observation layer used in retrospective evaluation. The system architecture is organized so that the spread model generates a daily forecast, the drone layer is assigned toward informative fire-front regions, and the collected observations are assimilated back into the model before the next forecast step.

This design is operationally motivated but evaluated here only in a retrospective replay setting: the CA forecast acts as the predictive backbone, while the simulated observation layer provides localized state corrections that can be examined without claiming historical live UAV deployment during the 2021 event.

The CA and MAS components are coupled through a lightweight binary assimilation module that applies explicit cell-level corrections to the current fire state. This configuration supports daily updates across a large raster domain while suppressing spurious ignitions, restoring missed burning cells, and improving delineation of the fire front.

## Retrospective Wildfire-Front Forecasting Cycle



**Fig 1. Closed-loop workflow of the retrospective CA+MAS wildfire forecasting framework.**

The sequence runs from multi-source data preparation and daily CA spread prediction to informative-region selection, drone-task allocation, targeted aerial observation, binary confirm-deny assimilation, and propagation of the corrected fire state to the next daily forecast step.

The cellular automaton generates a daily fire-front forecast that is used to identify informative regions for drone assignment. Observations collected by the drone layer are passed to the assimilation module, which updates the current model state before the next forecast step. This closed-loop sequence explicitly follows forecast, target selection, observation, state correction, and updated CA propagation. Fig 1 summarizes both the order of operations and the information flow among the CA, MAS, and assimilation modules.

## Data acquisition

The retrospective case study used a prepared daily geospatial dataset for the 2021 Dixie Fire in California, USA. In the data-preparation procedure, the study area was defined as a geographic bounding box with approximate limits  $[-121.5, 39.5, -120.5, 40.5]$ , and all static and time-varying inputs were assembled for this area on a common 100 m grid. The analyzed temporal window extended from 1 July 2021 to 30 September 2021, yielding 92 daily burned-area masks and matching daily meteorological fields.

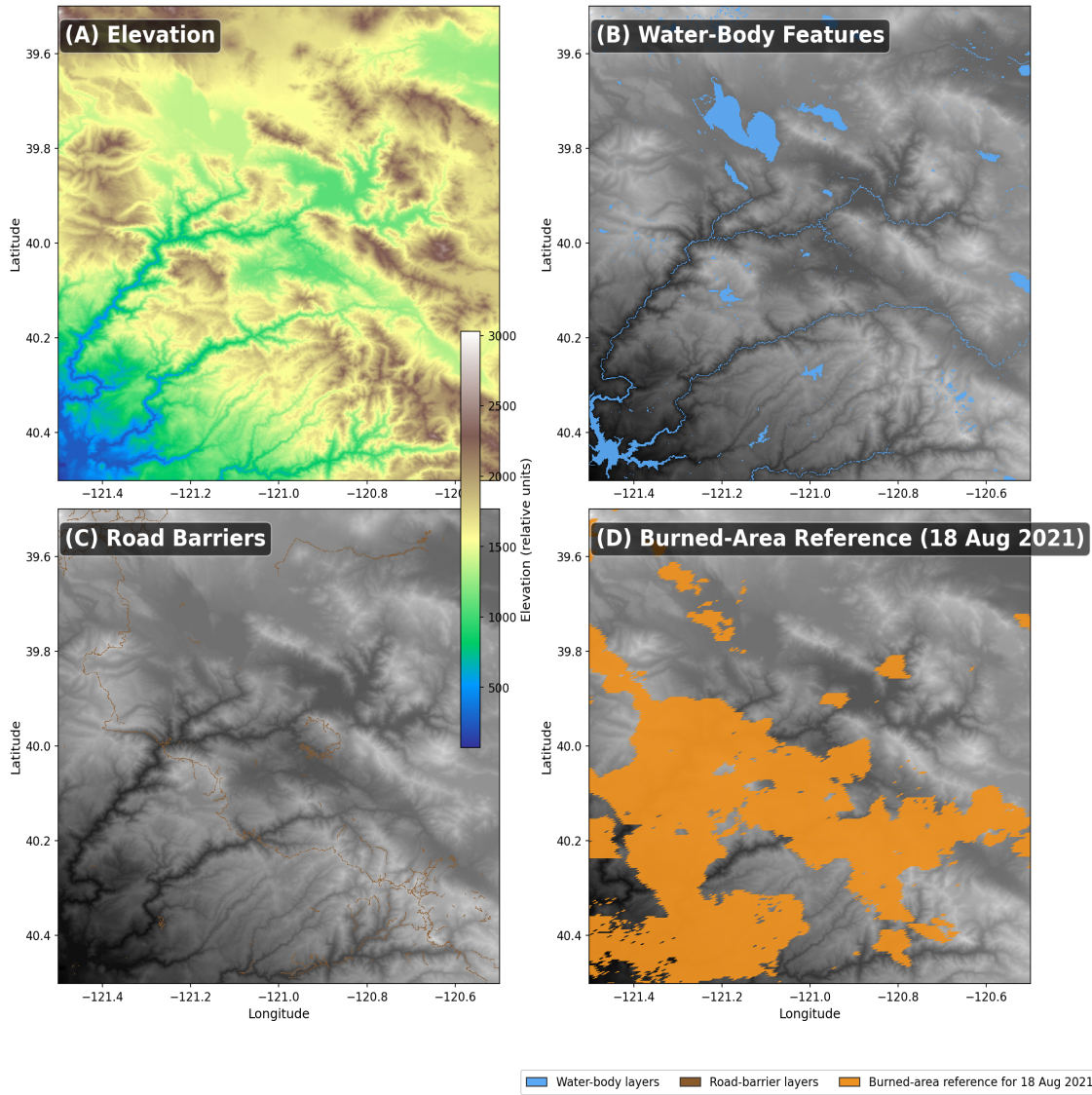
Static spatial inputs included digital elevation, slope, aspect, land cover, a land-cover-derived fuel proxy, median NDVI, and barrier layers representing hydrographic features. The prepared geospatial dataset includes DEM, slope, aspect, fuel\_model, landcover, ndvi\_median, barriers\_roads, and barriers\_water rasters. Where a dedicated fuel raster is unavailable, the analysis pipeline derives a simple fuel proxy from land-cover classes. In that sense, the fuel representation follows the same general idea as fuel-load mapping workflows that combine land-cover information with vegetation-condition indicators, even though the present study uses a lightweight case-study proxy rather than a separately trained fuel-map product.

Dynamic daily inputs consisted of MODIS burned-area rasters, VIIRS active-fire observations, and ERA5-Land-based meteorological forcing fields. In the analysis pipeline, the daily weather stack includes zonal and meridional 10 m wind components, wind speed, wind direction, 2 m air temperature, 2 m dewpoint temperature, vapor pressure deficit, relative humidity, and total precipitation. MODIS burned-area and VIIRS products are used as retrospective fire-observation layers, whereas ERA5-Land variables provide meteorological forcing rather than direct fire measurements. The geospatial preparation procedure therefore draws on SRTM terrain data, ESA WorldCover, MODIS MOD13Q1 vegetation indices, MODIS MCD64A1 burned-area maps, VIIRS VNP14A1 active-fire products, ERA5-Land reanalysis fields, and OpenStreetMap vector layers processed in QGIS [18–25].

For manuscript-level reproducibility, all layers were reprojected to the common working grid before model execution. Continuous rasters such as terrain, NDVI, and meteorological fields were resampled with bilinear interpolation, whereas categorical or binary layers such as land cover, burned masks, VIIRS detections, and barrier masks were preserved with nearest-neighbor resampling or direct rasterization. For ERA5-Land forcing, hourly temperature, dewpoint, and wind components were aggregated to daily means, while total precipitation was aggregated to a daily sum.

Fig 2 illustrates the integration of representative spatial inputs used to build the case-study grid. Panel A shows the digital elevation model used to derive terrain controls, Panel B shows the water-body features used in the barrier construction, Panel C shows the mapped road barriers, and Panel D shows the burned-area reference for 18 August 2021 used in the retrospective evaluation. Longitude and latitude axes are shown directly in the map frame, and the figure includes a terrain color scale together with legend entries for the binary overlay layers.

Taken together, these layers show how the DEM-based spatial framework is progressively complemented by water-body, road-barrier, vegetation, and retrospective fire-observation inputs before model execution.



**Fig 2. Representative spatial layers used to assemble the Dixie Fire case-study grid.** (A) Elevation from the DEM used to derive terrain controls with longitude and latitude axes and a DEM color scale. (B) Water-body features incorporated into the barrier construction. (C) Road barriers rasterized onto the common working grid. (D) Burned-area reference for 18 August 2021 used in the retrospective evaluation. The bottom legend identifies the binary overlay layers used in panels (B)–(D).

The resulting dataset is therefore a retrospective multi-source daily stack that combines terrain, vegetation, fuel, barrier, weather, and observational fire information for one case study. This stack forms the common input basis for both the baseline cellular automaton and the hybrid CA+MAS evaluation. Although the prepared dataset spans the full period from 1 July to 30 September 2021, the quantitative comparison reported in this manuscript is intentionally restricted to a verified 5-day mid-fire forecasting window from 14 to 18 August 2021, with 14 August used to initialize the forecast state and daily forecast skill reported for 15 to 18 August.

For clarity, Table 1 summarizes the datasets used to assemble the case-study stack, together with their native temporal and spatial resolutions and their roles in the retrospective evaluation.

**Table 1.** Data sources and roles in the retrospective CA+MAS case-study dataset.

| Dataset            | Source                           | Temporal resolution | Spatial resolution | Purpose                                         |
|--------------------|----------------------------------|---------------------|--------------------|-------------------------------------------------|
| DEM, Slope, Aspect | SRTM-derived terrain layers [18] | Static              | 30 to 100 m        | Topography                                      |
| Land Cover         | ESA WorldCover [19]              | Static              | 10 to 100 m        | Fuel proxy baseline                             |
| NDVI               | MODIS                            | 16-day              | 250 to 100 m       | Vegetation condition modifier                   |
| Burned Area        | MOD13Q1 [20]                     | composite           |                    |                                                 |
|                    | MODIS                            | Daily               | 500 to 100 m       | Retrospective burned-area reference             |
|                    | MCD64A1 [21]                     |                     |                    |                                                 |
| Active Fires       | VIIRS VNP14A1 [22]               | Daily               | 375 to 100 m       | Retrospective active-fire guidance              |
| Meteorology        | ERA5-Land [23]                   | Hourly to daily     | 9 km to 100 m      | Wind, temperature, humidity, VPD, precipitation |
| Barriers           | OpenStreetMap and QGIS [24, 25]  | Static              | Vector to 100 m    | Road and water-body barriers                    |

Because the subsequent Methods subsections refer to one verified August configuration rather than to an abstract parameter space, the key settings adopted in the reported experiment are listed explicitly in Table 2.

**Table 2.** Verified August parameter settings used in the retrospective CA+MAS evaluation.

| Parameter                     | Value used  | Description                                                     |
|-------------------------------|-------------|-----------------------------------------------------------------|
| Grid resolution               | 100 m       | Common raster grid used for all static and daily layers.        |
| Grid size                     | 1115 × 1115 | DEM-referenced simulation domain.                               |
| Time step                     | 1 day       | CA update interval and forecast step.                           |
| Base spread probability $P_0$ | 0.15        | Baseline ignition probability used in the CA component.         |
| Number of drones              | 5           | Simulated UAV agents in the verified August experiment.         |
| Sensing radius                | 8 px        | Radius of the UAV observation footprint.                        |
| Movement speed                | 35 px/step  | Maximum UAV movement per intra-day step.                        |
| Intra-day update count        | 24          | Number of UAV movement and scan steps per simulated day.        |
| Burned assimilation radius    | 2 px        | Neighborhood reinforcement radius around observed burned cells. |
| Clear assimilation radius     | 2 px        | Neighborhood suppression radius around observed clear cells.    |

The following subsections describe how these inputs and parameter settings enter the preprocessing, spread modeling, observation, and assimilation stages of the computational experiment.

## Preprocessing

Preprocessing was designed to place all inputs on a common DEM-referenced raster grid before model execution. The reference DEM grid in the prepared dataset has dimensions 1115 x 1115 cells, and all static layers are matched to this shape in the computational procedure. This alignment step ensures that terrain, fuel, barriers, and meteorological forcing fields can be used consistently by the cellular automaton and the hybrid observation-assimilation pipeline.



The preprocessing sequence consisted of four main operations. First, source rasters were loaded from the prepared geospatial dataset and truncated or shape-matched to the DEM reference grid when necessary. Second, NDVI values were clipped to the unit interval before use as a vegetation-condition modifier. Third, the two barrier rasters were binarized and merged into one combined barrier mask. After correction of the previously invalid full-coverage masks, the barrier layers occupied only a minority of pixels and therefore behaved as localized spatial constraints rather than domain-wide exclusions. Fourth, burned-area rasters were converted to daily binary masks, with positive values denoting cells classified as burned on that day.

In compact form, the preprocessing rules used in the analysis pipeline can be written as

$$\text{NDVI}^*(i, j) = \min(1, \max(0, \text{NDVI}(i, j))) \quad (1)$$

$$M_{\text{barrier}}(i, j) = \min(1, M_{\text{road}}(i, j) + M_{\text{water}}(i, j)) \quad (2)$$

and

$$F_t^{\text{obs}}(i, j) = \mathbf{1}_{\{B_t(i, j) > 0\}} \quad (3)$$

where Eqs. 1–3 define the three core preprocessing transformations used before model execution. Here  $\text{NDVI}^*$  is the clipped vegetation-condition layer,  $M_{\text{road}}$  and  $M_{\text{water}}$  are the binary road and water masks rasterized on the DEM grid,  $M_{\text{barrier}}$  is the merged barrier mask, and  $F_t^{\text{obs}}$  is the daily binary burned-area observation derived from the retrospective burned raster  $B_t$ .

The daily weather stack was indexed by date and linked to the burned-area sequence so that each simulation step used a matching set of meteorological drivers. If an expected daily weather raster was missing, the preprocessing procedure filled that variable with zeros for the corresponding day; however, in the assembled Dixie Fire dataset all required weather variables were available for the full 92-day analysis window.

This preprocessing procedure yields a unified daily data cube in which every layer is spatially aligned and available at the same simulation resolution. For the verified 5-day mid-fire experiment reported in this manuscript, the corrected combined barrier mask was enabled as a spatial constraint in both the standalone CA baseline and the CA component of the hybrid model. The standardized stack is then passed to the stochastic cellular automaton, the simulated UAV observation module, and the subsequent binary assimilation step.

## Cellular Automaton

The standalone spread component is a stochastic binary cellular automaton operating on the daily burned or unburned raster state. Let  $F_t(i, j) \in \{0, 1\}$  denote the fire state of cell  $(i, j)$  on day  $t$ , where 1 indicates burned and 0 indicates unburned. A candidate cell is eligible for ignition only if it is currently unburned and at least one of its eight neighboring cells is burning.

For eligible cells, the ignition probability is computed as one base probability multiplied by terrain, fuel, weather, barrier, and directional correction factors, as summarized in Eq. 4:

$$P_t(i, j) = P_0 S(i, j) W(i, j) U(i, j) T(i, j) V(i, j) H(i, j) R(i, j) B(i, j) D(i, j) G_t(i, j) \quad (4)$$

where  $P_0 = 0.15$  is the base spread probability used in the model. The factors are easier to interpret in groups. Terrain and fuel enter through the slope factor  $S(i, j) = 1 + \text{slope}(i, j)/100$  and the normalized fuel factor  $U(i, j) = \text{fuel}(i, j) / \max(\text{fuel})$ , so steeper slopes and more combustible local fuel increase the chance of ignition. Wind and temperature act as spread amplifiers through  $W(i, j) = 1 + \text{wind.speed}(i, j)/10$  and  $T(i, j) = 1 + 0.005 \max(T_c(i, j) - 20, 0)$ , which raise the transition probability under stronger winds and warmer near-surface air. Fuel dryness and moisture limitation are represented by  $V(i, j) = 1 + 0.02 \max(\text{VPD}(i, j), 0)$ ,  $H(i, j) = 1 / (1 + 0.01 \max(\text{RH}(i, j), 0))$ , and  $R(i, j) = \exp(-2 \max(\text{TP}(i, j), 0))$ , so high vapor pressure deficit promotes spread whereas high humidity and precipitation suppress it.

The remaining terms control where spread is allowed and in which direction it is favored. The barrier term  $B(i, j)$  equals 0 on cells covered by the combined road or water barrier mask when barriers are enabled

and 1 otherwise. The wind-direction term  $D(i, j)$  increases the probability by up to 30% when the local wind direction is aligned with the neighborhood-derived direction of incoming spread. An optional multiplicative spread-bias term may also be introduced:

$$G_t(i, j) = \max(0, 1 + g_t(i, j)) \quad (5)$$

where Eq. 5 defines the optional multiplicative bias term. The quantity  $g_t(i, j)$  is a bounded next-day bias field derived from observed growth cells, observed frontier burning, and observed clear-growth regions. When the spread-bias option is disabled,  $g_t(i, j) = 0$  and the CA reduces to the base multiplicative form above.

After multiplication,  $P_t(i, j)$  is clipped to the interval  $[0, 1]$  and the state update is sampled from a Bernoulli trial, so ignition occurs when a uniform random draw is smaller than  $P_t(i, j)$ . Burned-area rasters were binarized by assigning all positive raster values to the burned class, and no explicit lightning-ignition term is included in this study. The daily state is therefore driven by a simple sequence: a cell must have a burning neighbor, the local multiplicative factors determine its daily ignition probability, and a Bernoulli draw converts that probability into the next binary burned or unburned state. This keeps the CA mathematically explicit while preserving a direct physical reading of each factor.

## Multi-Agent System

The multi-agent subsystem is implemented as a simulated observation layer that operates on top of the daily cellular automaton forecast. In this framework, the number of drones, sensing radius, movement speed, and intra-day update count are configurable runtime parameters rather than fixed properties of the method. Each agent stores its current grid position, sensing radius, movement speed, observed cells, assigned region, and a day-specific trajectory history that is later used for reporting and visualization.

These agents should be understood as a proxy for recurring daily aerial observations within a retrospective forecast-update cycle rather than as a literal representation of uninterrupted multi-day UAV flight. In this formulation, each simulated day corresponds to one observation-update window in which targeted aerial information becomes available before the next daily forecast step.

For agent  $a$ , the simulated drone state is represented as Eq. 6:

$$s_t^{(a)} = (y_t^{(a)}, x_t^{(a)}, r^{(a)}, v^{(a)}) \quad (6)$$

where  $(y_t^{(a)}, x_t^{(a)})$  is the current grid position,  $r^{(a)}$  is the sensing radius, and  $v^{(a)}$  is the maximum movement speed in pixels per intra-day step. If  $\mathbf{p}_t^{(a)}$  denotes the current position and  $\mathbf{g}_t^{(a)}$  the active target point, then the implemented movement rule is the bounded directed step given in Eq. 7:

$$\Delta \mathbf{p}_t^{(a)} = \min(v^{(a)}, \|\mathbf{g}_t^{(a)} - \mathbf{p}_t^{(a)}\|_2) \frac{\mathbf{g}_t^{(a)} - \mathbf{p}_t^{(a)}}{\|\mathbf{g}_t^{(a)} - \mathbf{p}_t^{(a)}\|_2} \quad (7)$$

followed by clipping to the valid raster domain. When no target is active, the fallback behavior is a bounded random move on the same grid.

At the start of a run, drones are initialized on burning pixels from the first available fire-state estimate; if no burning pixels are present, they are initialized at random grid locations. On each subsequent day, the simulation design derives candidate observation regions from the predicted growth field and the forecast fire front, aggregates these cells into spatial clusters, and ranks the resulting clusters to preserve both informational value and spatial diversity. Drone-to-region allocation is then solved as a distance-minimization problem using the Hungarian algorithm for linear sum assignment [26]; a greedy distance-based rule can be used as a fallback implementation.

The corresponding pairwise assignment cost is the Euclidean distance between drone and target positions, as given by Eq. 8,

$$c_{ab} = \|\mathbf{p}_t^{(a)} - \mathbf{g}_t^{(b)}\|_2, \quad (8)$$

and the Hungarian step selects a one-to-one assignment that minimizes the total distance across the active drone and target sets. When SciPy is unavailable, the analysis pipeline falls back to a greedy distance-based matching rule.

Drone motion is modeled in grid coordinates. When a drone has an assigned observation region, it follows a short intra-cluster patrol defined by representative waypoints and advances toward the active waypoint by at most its configured speed in pixels per intra-day step. When no explicit assignment is active, the drone moves toward the nearest currently unexplored informative cell. If no such cell is available, the drone performs a bounded random move within the grid.

Observations are collected over the swept corridor traced by each drone during intra-day motion rather than only at waypoint positions. In practice, the observation set is represented as the union of local circular sensing footprints sampled along the traversed path. Newly observed burning cells are written to a shared global observation map that aggregates all detections made by the simulated agents during the current daily update cycle.

Accordingly, the observed set for drone  $a$  over one daily update window can be summarized by Eq. 9:

$$O_t^{(a)} = \bigcup_{k=0}^{K_t^{(a)}} \left\{ (i, j) : \left\| (i, j) - \mathbf{p}_{t,k}^{(a)} \right\|_2 \leq r^{(a)} \right\} \quad (9)$$

where  $\mathbf{p}_{t,k}^{(a)}$  are the sampled intra-day positions along the path of the drone. The global observation support used by the assimilation step is then  $K_t = \bigcup_a O_t^{(a)}$ .

In this formulation, the MAS does not replace the spread model and does not perform obstacle-aware route planning. Its role is to concentrate simulated observations on informative fire-front regions and to provide a compact, reproducible mechanism for retrospective sensing within the hybrid CA+MAS framework.

For reproducibility, the framework exposes the CA base spread probability, drone count, sensing radius, movement speed, intra-day update count, targeting strategy, assignment rule, spread-bias toggle, and assimilation radii as explicit runtime parameters. The manuscript results reported here correspond to one verified August parameter set, summarized in Table 2, rather than to generic default settings.

## Data Assimilation

Data assimilation is implemented as a lightweight binary state-correction step rather than as an Ensemble Kalman Filter or a Particle Filter [27, 28]. Because the fire map is represented as a binary burned or unburned raster, the assimilation logic operates directly on binary cell states and therefore remains consistent with the underlying cellular automaton representation.

After the intra-day drone motion and scanning steps are completed, the evaluation protocol copies the current CA prediction and applies a confirm-deny update on all cells present in the shared global observation map. For each observed location, the corrected fire map is set to 1 when the retrospective burned reference indicates fire presence and to 0 otherwise. In this way, confirmed ignitions are inserted into the model state and false activations are removed before the next daily CA spread step is executed.

Let  $F_t^{\text{pred}}$  denote the daily CA prediction before assimilation, let  $F_t^{\text{ref}}$  denote the retrospective burned reference used during replay, and let  $K_t$  denote the union of cells observed by all drones during that day. The implemented binary correction operator is given in Eq. 10:

$$\tilde{F}_t(i, j) = \begin{cases} 1, & (i, j) \in K_t \text{ and } F_t^{\text{ref}}(i, j) = 1, \\ 0, & (i, j) \in K_t \text{ and } F_t^{\text{ref}}(i, j) = 0, \\ F_t^{\text{pred}}(i, j), & (i, j) \notin K_t. \end{cases} \quad (10)$$

Thus, observation-supported burned cells are confirmed, observation-supported false positives are removed, and unobserved cells retain the CA state predicted for that day.

In the present formulation, this direct cellwise update is then extended to small local neighborhoods around observed burned and observed clear cells using configurable dilation radii. Burned observations therefore reinforce nearby frontier and growth pixels, whereas clear observations suppress nearby false

activations. The core confirm-deny logic of Eq. 10 is preserved, but the effective update is slightly more spatially permissive than a purely pointwise overwrite.

The corrected map is then passed to the CA spread operator to generate the next hybrid forecast. In parallel, the evaluation procedure also propagates an uncorrected CA state for comparison purposes. This produces a day-by-day paired evaluation in which the hybrid forecast and the baseline CA forecast can be scored against the same retrospective burned-area target while sharing the same meteorological forcing and barrier configuration.

If  $\Phi(\cdot; \theta_t)$  denotes the CA transition operator under the weather and static inputs for day  $t$ , then the hybrid and baseline forecast branches can be summarized as in Eq. 11:

$$F_{t+1}^{\text{hyb}} = \Phi(\tilde{F}_t; \theta_t), \quad F_{t+1}^{\text{base}} = \Phi(F_t^{\text{pred}}; \theta_t) \quad (11)$$

This notation makes explicit that the only difference between the two branches is the insertion of the binary observation-correction step before the next daily CA propagation.

This assimilation design has two practical consequences. First, it is computationally lightweight because it avoids solving any high-dimensional inverse problem. Second, it remains interpretable because every correction corresponds to an explicit cell-level observation supplied by the simulated drone layer. For the present retrospective case study, this simple binary confirm-deny rule is sufficient to test whether targeted observations improve short-horizon wildfire-front reconstruction.

The same observed footprint can also be used to blend the daily weather fields locally before the next forecast step. If  $X_t^{\text{prior}}$  and  $X_t^{\text{sens}}$  denote the prior and sensor-derived weather rasters for a given variable and  $\Omega_t^{\text{obs}}$  denotes the dilated observed footprint, then the local weather update is given by Eq. 12:

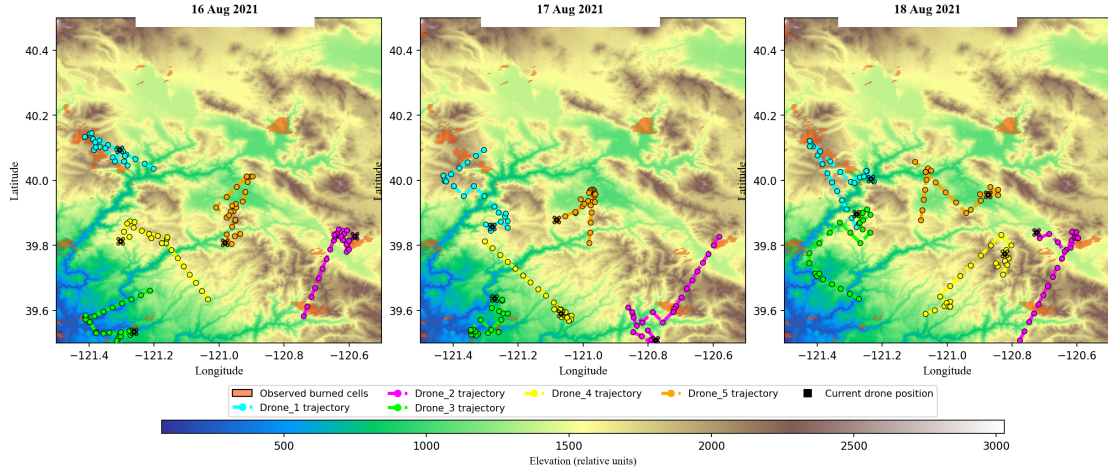
$$X_t^{\text{assim}}(i, j) = \begin{cases} (1 - \lambda_w)X_t^{\text{prior}}(i, j) + \lambda_w X_t^{\text{sens}}(i, j), & (i, j) \in \Omega_t^{\text{obs}}, \\ X_t^{\text{prior}}(i, j), & (i, j) \notin \Omega_t^{\text{obs}}, \end{cases} \quad (12)$$

with blend weight  $\lambda_w \in [0, 1]$ . This local meteorological update is auxiliary: it complements rather than replaces the primary binary confirm-deny correction of the fire-state raster.

## Results

The study results demonstrate that the proposed method improves short-horizon fire-front reconstruction relative to the baseline cellular automaton. In the retrospective hybrid framework, drone trajectories remain concentrated near active and previously unobserved burning regions, which increases the coverage of informative fire-front segments during the simulated observation stage (Fig 3).

Representative simulated drone trajectories during the mid-fire period



**Fig 3. Composite view of representative simulated drone trajectories during the middle portion of the retrospective fire period.** (a) 16 August 2021, (b) 17 August 2021, and (c) 18 August 2021. Orange polygons denote observed burned cells, colored dashed lines denote individual drone trajectories, square markers denote current drone positions, and the DEM background provides terrain context for each daily observation cycle.

Across successive daily update cycles, the simulated observation layer remains concentrated near active fire-front segments, providing dense localized coverage where state correction is most informative. This adaptive sensing pattern is visible in the hybrid configuration. The later dates shown in Fig 3 also provide a clearer visual demonstration of how the agents redistribute as the front evolves across the verified window.

To quantitatively evaluate the contribution of the multi-agent subsystem to modeling accuracy, a comparative analysis was conducted between the baseline CA and the hybrid CA+MAS architecture incorporating drone observations.

The simulated observation policy redirects sensing effort across the retrospective experiment as fire-front geometry changes, while coverage remains concentrated near active segments.

For the adopted 5-day evaluation window from 14 to 18 August 2021, with corrected road and water barrier masks enabled in the CA component, the standalone baseline CA achieved a mean Intersection over Union (IoU) of 0.6664, whereas the hybrid CA+MAS configuration reached 0.6992. This corresponds to an absolute IoU gain of 0.0328, indicating that the assimilation of simulated aerial observations improved short-horizon wildfire-front reconstruction over the barrier-aware baseline forecast (Table 3).

**Table 3.** Comparative summary of mean Intersection over Union (IoU) for the standalone CA baseline and the CA+MAS hybrid configuration.

| Model Variant                | Mean IoU | Absolute Gain | Relative Gain |
|------------------------------|----------|---------------|---------------|
| CA (Cellular Automaton only) | 0.6664   | —             | —             |
| CA+MAS (with drone feedback) | 0.6992   | +0.0328       | +4.92%        |

Because the revised comparison is intentionally limited to the verified 5-day experiment, we report only those summary metrics that are directly reproduced by the reproducible evaluation protocol. In addition to the higher mean IoU, the hybrid run yielded a mean F1 score of 0.8229 compared with 0.7998 for the standalone baseline CA, which is consistent with the visual improvement shown in the comparative figures.

Across the evaluated forecast days, the hybrid CA+MAS system yields higher IoU and F1 than the standalone cellular automaton, as summarized in Table 4. The mean IoU improvement for this 5-day experiment is +0.0328 (from 0.6664 to 0.6992), corresponding to a relative increase of +4.92 %, while the mean F1 score increases from 0.7998 to 0.8229. Positive daily IoU gains are observed on every evaluated day of the verified window.

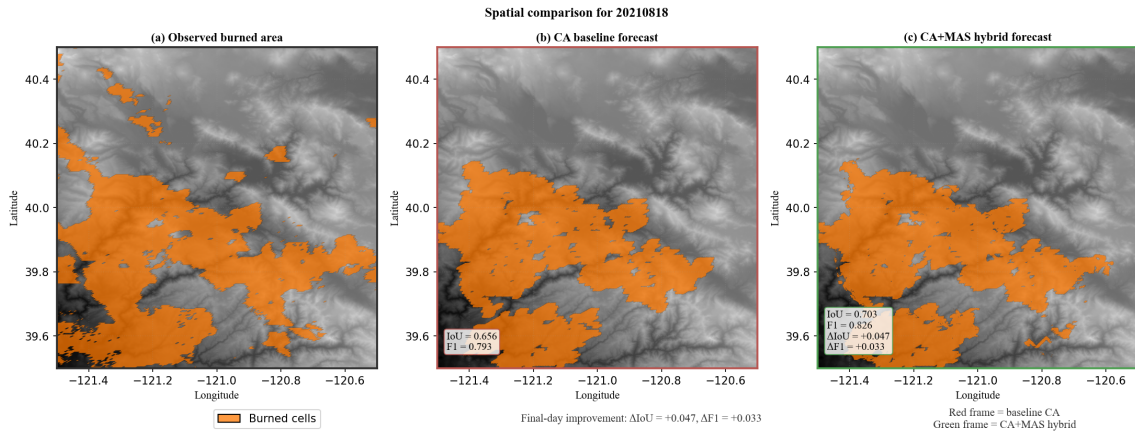
**Table 4.** Daily and mean performance comparison between the standalone CA baseline and the CA+MAS hybrid configuration over the verified evaluation window. The 14 August 2021 raster is used as the initialization state; therefore, daily forecast metrics are reported for 15 to 18 August 2021.

| Date        | IoU CA        | IoU Hybrid    | $\Delta$ IoU   | F1 CA         | F1 Hybrid     | $\Delta$ F1    |
|-------------|---------------|---------------|----------------|---------------|---------------|----------------|
| 20210815    | 0.6670        | 0.6818        | +0.0148        | 0.8003        | 0.8108        | +0.0105        |
| 20210816    | 0.6731        | 0.7037        | +0.0305        | 0.8046        | 0.8261        | +0.0214        |
| 20210817    | 0.6690        | 0.7080        | +0.0390        | 0.8017        | 0.8290        | +0.0274        |
| 20210818    | 0.6563        | 0.7032        | +0.0470        | 0.7925        | 0.8258        | +0.0333        |
| <b>Mean</b> | <b>0.6664</b> | <b>0.6992</b> | <b>+0.0328</b> | <b>0.7998</b> | <b>0.8229</b> | <b>+0.0232</b> |

Fig 4 presents a spatial comparison for the final evaluated day of the verified window (18 August 2021), showing:

- (a) the observed burned-area reference;
- (b) the forecast produced by the standalone cellular automaton (CA) baseline; and
- (c) the forecast generated by the hybrid CA+MAS model after correction with the simulated daily aerial observation layer.

The visual comparison places the observed burned-area reference, the standalone CA baseline forecast, and the hybrid CA+MAS forecast side by side for the same final evaluated day.



**Fig 4. Spatial comparison on 18 August 2021 between the observed burned-area reference, the standalone CA baseline forecast, and the CA+MAS hybrid forecast corrected by the simulated daily aerial observation layer.** The numerical annotations within panels (b) and (c) report final-day IoU and F1 values for 18 August 2021, whereas the summary values discussed in Table 3 and Table 4 correspond to means aggregated across the full verified evaluation window.

Taken together, the spatial comparison and tabulated metrics describe the same verified short-horizon evaluation window.

## Discussion

The verified 5-day mid-fire experiment indicates that the hybrid CA–MAS configuration improves short-horizon wildfire-front reconstruction because the forecast state is corrected before errors can accumulate across successive daily steps. In the present framework, the gain does not arise from a more complex spread model, but from inserting targeted simulated observations into the daily forecast-update cycle. This makes the observed improvement mechanistically consistent with the simulation design: the CA provides an interpretable baseline forecast, the MAS concentrates attention on informative front segments, and the binary assimilation rule removes false activations while restoring missed burned cells.

Classical cellular automata models [10, 11] lack intrinsic mechanisms for adapting to changing conditions once a forecast has been initialized. In contrast, the proposed CA+MAS system incorporates an explicit forecast-observe-correct loop driven by targeted simulated UAV observations. This architecture improves the suitability of the model for short-horizon updating when wind, fuel, and vegetation conditions cause the fire front to evolve rapidly.

In comparison with physical-mathematical advection-diffusion-reaction (ADR) models [12], the CA+MAS framework operates on aggregated input layers and is readily scalable to large spatial domains. This contrast is relevant to the present results because the observed IoU improvement was obtained without introducing a high-parameter physical solver; instead, the improvement emerged from a computationally lightweight correction loop applied to the same daily forcing and barrier-aware CA backbone. The current case study therefore suggests that, for short retrospective windows, targeted observation and state correction can yield useful accuracy gains even within a simpler spread-modeling framework.

Deep neural network architectures, including CNNs, convolutional long short-term memory (ConvLSTM) networks, and transformers, have demonstrated high predictive accuracy for fire dynamics based on satellite observations [6, 14]. However, these approaches typically require large volumes of labeled training data, offer limited interpretability, and exhibit reduced generalizability when applied to previously unseen regions without additional retraining. In contrast, the CA+MAS framework retains physically interpretable parameters, operates independently of training datasets, and supports explicit state updating through the simulated observation layer.

A key distinguishing feature of the hybrid architecture is the explicit use of targeted observations and shared observation-map assimilation within a lightweight CA framework. Although wildfire forecasting studies have also explored latent data assimilation and satellite-based fire-growth updates [7–9], the present model avoids solving a high-dimensional filtering problem. Instead, it uses a simulated daily aerial observation layer to correct missed burning cells and suppress false activations before the next spread step. This design helps explain why the model performs best in a short-horizon retrospective setting: corrections are local, interpretable, and directly tied to the same cells that drive the next daily forecast update.

The modular architecture of the system enables straightforward expansion of its functional capabilities. Potential extensions include the incorporation of multi-agent reinforcement learning algorithms for autonomous swarm coordination, the integration of higher-resolution meteorological models, and embedding within broader situational-awareness pipelines. These extensions are motivated by the current findings rather than independent of them: because the verified case study supports the value of targeted observation and explicit state correction, future work should focus on improving where, when, and how those observations are generated rather than on broadening the claims of the present study. In its current form, the CA+MAS framework should be understood as an interpretable and extensible prototype for wildfire monitoring and forecasting rather than as a fully operational end-to-end deployment.

## Limitations

Several limitations should be stated explicitly. First, the present evaluation is based on a single retrospective case study and a verified 5-day mid-fire window, so the reported gains should not yet be treated as evidence of broad cross-fire generalization. Second, the aerial component is represented by simulated UAV observations rather than by a historical deployment log, which means that sensing coverage, movement, and correction timing were assessed within a computational experiment rather than under operational field constraints. Third, the data-assimilation module is intentionally simple: it applies explicit binary state correction and local neighborhood dilation, but it does not estimate posterior uncertainty or solve a higher-dimensional filtering problem. Fourth, although the verified August parameter set is now reported explicitly, the present manuscript does not claim that this parameterization is optimal outside the evaluated setting. These limits do not invalidate the present findings, but they define the scope within which the reported improvements should be interpreted.

## Conclusions and future directions

A hybrid architecture for wildfire modeling and forecasting has been developed by integrating a cellular automaton (CA) with a simulated drone-based multi-agent system (MAS). The model operates through an adaptive closed-loop sequence comprising CA simulation, MAS-driven observation, data assimilation, and state update. In the verified 5-day evaluation window from 14 to 18 August 2021, with corrected barrier masks enabled in the CA component, this integrated approach increased mean IoU from 0.6664 for the standalone baseline CA to 0.6992 for the hybrid configuration, while the mean F1 score increased from 0.7998 to 0.8229. These results indicate that spatially targeted recurrent aerial observations can improve short-horizon wildfire-front forecasting.

The model offers high interpretability, linear scalability, and a comparatively lightweight update mechanism. Compared with alternatives such as FARSITE, deep learning-based approaches, and EnKF-style filters, it requires fewer computational resources, adapts its state through explicit observations, and preserves physical consistency in the forecast-update loop.

The present retrospective results suggest that the developed approach can inform future decision-support and situational-awareness frameworks for wildfire management. Its modularity and transparency make it suitable for further experimentation across additional regions and scenario types, provided that the present study is understood as a retrospective evaluation of simulated daily aerial observations rather than as a direct operational deployment study.

Future work may extend the hybrid architecture through the integration of machine learning and reinforcement learning techniques. In particular, the multi-agent UAV subsystem may evolve from deterministic task allocation toward adaptive behaviors derived from multi-agent reinforcement learning (MARL) algorithms, enabling drones to autonomously optimize routes, observation priorities, and operational strategies under uncertainty. Another avenue involves employing ML models for the dynamic calibration of cellular automaton parameters, including fuel-proxy scaling, wind and humidity adjustment terms, precipitation damping, and daily state-transition probabilities. These enhancements may strengthen forecast robustness and support the development of a more adaptive forecast–observe–learn cycle, thereby improving predictive accuracy and the flexibility of future wildfire-modeling frameworks.

## Data Availability Statement

The original input datasets used in this study are publicly available from the repositories listed in Table 1. Processed intermediate files are available from the corresponding author upon reasonable request.

## Code Availability Statement

The simulation scripts used to generate the reported results are available from the corresponding author upon reasonable request.

## Funding Statement

The authors received no specific funding for this work.

## Competing Interests

The authors declare that they have no competing interests.

## Acknowledgments

The authors have no additional acknowledgments to report.



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